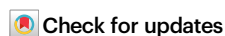


Understanding epistatic networks in the B1 β -lactamases through coevolutionary statistical modeling and deep mutational scanning

Received: 6 February 2024

Accepted: 16 September 2024

Published online: 30 September 2024

J. Z. Chen¹, M. Bisardi^{2,3}, D. Lee¹, S. Cotogno^{2,3}, F. Zamponi^{2,4}, M. Weigt³ & N. Tokuriki¹✉

Throughout evolution, protein families undergo substantial sequence divergence while preserving structure and function. Although most mutations are deleterious, evolution can explore sequence space via epistatic networks of intramolecular interactions that alleviate the harmful mutations. However, comprehensive analysis of such epistatic networks across protein families remains limited. Thus, we conduct a family wide analysis of the B1 metallo- β -lactamases, combining experiments (deep mutational scanning, DMS) on two distant homologs (NDM-1 and VIM-2) and computational analyses (in silico DMS based on Direct Coupling Analysis, DCA) of 100 homologs. The methods jointly reveal and quantify prevalent epistasis, as $\sim 1/3$ rd of equivalent mutations are epistatic in DMS. From DCA, half of the positions have a >6.5 fold difference in effective number of tolerated mutations across the entire family. Notably, both methods locate residues with the strongest epistasis in regions of intermediate residue burial, suggesting a balance of residue packing and mutational freedom in forming epistatic networks. We identify entrenched WT residues between NDM-1 and VIM-2 in DMS, which display statistically distinct behaviors in DCA from non-entrenched residues. Entrenched residues are not easily compensated by changes in single nearby interactions, reinforcing existing findings where a complex epistatic network compounds smaller effects from many interacting residues.

The comprehension of protein sequence-structure-function relationship is essential to the understanding of biology and biotechnological applications. Evolving over billions of years, protein sequences spread among diverse organisms accumulating mutations and diverging into families of homologs often sharing $<25\%$ sequence identity^{1,2}. Evolution navigates this vast sequence space while

maintaining protein function and avoiding ‘fitness valleys’ ($\sim 35\text{--}70\%$ of mutations are deleterious)^{3–12}. Intriguingly, many deleterious mutations in one protein can be observed in other homologs in the context of many other mutations¹³. Networks of intramolecular interactions (epistatic networks) aid fitness landscape exploration by alleviating the effects of deleterious mutations (via contingency or

¹Michael Smith Laboratories, University of British Columbia, Vancouver, BC, Canada. ²Laboratoire de Physique de l’Ecole Normale Supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université de Paris, F-75005 Paris, France. ³Sorbonne Université, CNRS, Institut de Biologie Paris Seine, Biologie Computationnelle et Quantitative LCQB, F-75005 Paris, France. ⁴Dipartimento di Fisica, Sapienza Università di Roma, I-00185 Rome, Italy.

✉ e-mail: tokuriki@msl.ubc.ca

compensation), but may cause other mutations to become deleterious in turn (via entrenchment)^{14–16}. Hence, complex intramolecular epistasis causes differential mutational behaviors between homologs.

The evolution of natural proteins thus prompts many important and unanswered questions. What causes function-destroying mutations in one sequence to be fixed in the sequence of homologs? How often are such heterogeneities in mutational effects encountered in a protein family? Can these patterns be modeled and predicted? Such epistatic networks are key to understanding sequence-structure-function relationships. Recent technological advances allow us to study epistatic networks based on the coevolution within a homologous family (genomics¹⁷), or by experimentally measured mutational behaviors (biochemical¹⁸). Statistical models trained on multiple sequence alignments can capture coevolutionary patterns^{19,20} and generate artificial sequences that respect the statistics of protein families^{19,21–25}. Such data-driven computational approaches are able to design functional protein sequences unseen in nature^{26–28}, but to what extent and accuracy such models capture specific epistatic interactions remains unknown. Experimentally, epistatic networks are detected via mutational incompatibilities between different genetic backgrounds. Such experiments provide an in-depth understanding of specific biochemical and biophysical interactions within proteins. However, experiments of family-wide functional constraints that compare residue-level mutational epistasis across homologs remain scarce, as only a small number of protein families have been studied¹³, including a few large-scale datasets^{29–35}. Notably, the statistical and experimental approaches provide independent and complementary information that can be combined to better understand epistatic networks, but no previous study has performed such integration. The computational power of generative models allows for analysis of many more sequences than is possible experimentally, and provides a global picture across the entire protein family. Meanwhile, the experimental data of local sequences space from DMS provides detailed observations specific to each homolog.

In this study, we employ the two approaches on the class B1 metallo- β -lactamases (MBL), a long-evolving, highly diversified antibiotic degrading enzyme family³⁶. In particular, we explore the compounded epistatic effects of very distant homologs (~70% sequence divergence) to interrogate the effects of epistasis across the scale of natural evolution. While studies of systematic combinations of lower order epistasis can reveal more detailed mechanisms^{37–39}, the compounding of epistatic effects is complex and extrapolation using sparse lower order epistasis would likely fail to capture epistatic outcomes at such large evolutionary distances^{35,40}. Furthermore, systematic studies of epistasis between highly diverged homologs remain rare^{30,35}. Using Direct Coupling Analysis (DCA)¹⁹, we analyze the coevolutionary signatures of the entire B1 family to probe global epistatic patterns by accumulating statistics over many homologs. We also perform deep mutational scanning (DMS) on two distant homologs, NDM-1 and VIM-2 (~30% sequence identity), in order to experimentally measure protein-specific mutational trends and incompatibilities. Thus, we have the opportunity to compare the results between methods and learn how each can strengthen and explain the other. We find general consensus between the two methods on the prevalence and strength of epistasis, and highlight cases where complementary information provides insight beyond that of a single method. Trends of mutational behavior emerge in 3D structures, but the exact intramolecular network appears much more complex upon further testing, as predicted by DCA and in line with previous findings. Overall, the combination of DCA and DMS allows us to greatly enhance our understanding of epistatic networks.

Results

Co-evolutionary model of the B1 family reveals sequence-specific mutational heterogeneities

Class B1 metallo- β -lactamases are a highly diversified family of enzymes (as low as 20% identity within the family) that degrade β -lactam antibiotics^{1,41,42}, a function for which there is likely a long evolutionary history^{36,43}. The homologs in this family have become highly diversified whilst retaining common functions (e.g., resistance to the same antibiotics¹), making this family an excellent model for epistatic networks, as the different homologs likely formed different sets of intramolecular interactions throughout their evolution. We collected all B1 MBL sequences (5516 total, 4046 non-redundant) (Supplementary Data 1), from Interpro⁴⁴, BLDB⁴⁵, and the Joint Genomics Institute IMG/M metagenomics database⁴⁶ (methods). Then, we inferred a global statistical model of the family using DCA (methods) that captures the coevolutionary relationships (direct couplings) between all residue pairs, and the residue conservation (fields). The model can assign a probability $P(\text{sequence}) \propto \exp\{-E(\text{sequence})\}$ to any sequence, proportional to the likelihood of being a functional MBL. To score single mutations, we take the difference in the model's "energy" (E) between the mutant at a position and the wild-type (WT), as $\Delta E = -\log[P(\text{mut}, \text{pos})/P(\text{WT})]$ ⁴⁷. Thus, the effect of any amino acid substitutions in any background is computable, with sequence context accounted for via coevolutionary couplings.

To probe the effects of epistatic networks across the B1 family, we first calculated ΔE for all single mutants of 100 diverse homologs (~70% sequence divergence, Supplementary Fig. 1a, methods). Then, for each individual position inside each homolog, we predict its mutational tolerance as the Shannon Entropy of all amino acid probabilities $-\exp(-\Delta E)$, also called context-dependent entropy (CDE)^{48,49} (Supplementary Data 2). This mutational tolerance is interpretable as the base-2 logarithm of the effective number of tolerated amino acids at a position in a specific WT background, where 0 means one (2^0 , the WT) residue is tolerated, and 4.3 means all twenty ($2^{4.3}$) amino acids are equally tolerated. Since the CDE is calculated from probabilities of all twenty amino acids at a position, the mutational tolerance is independent of the starting WT residue, and any difference between homologs in the CDE at any given position is necessarily a result of epistasis. The resulting mutational tolerance at each aligned position of the 100 homologs is shown Fig. 1b. When examining positions across all homologs there are highly constrained regions in terms of mutations, but also others that are highly tolerant. Between individual sequences, the relationship between sequence divergence and mutational tolerance across all positions can be quantified (Supplementary Fig. 1b). As expected there is a decreasing trend for similarity in sequence-level mutational tolerance with increased divergence, with a notable sharp drop around ~70% divergence. At a per position level, we can also observe the specific differences in mutational tolerance across the 100 homologs (Supplementary Fig. 1c). Hence, this method allows us to identify specific patterns of mutational behaviors in a large number of homologs.

There is substantial heterogeneity in behaviors across the 100 homologs, shown by the variability (interquartile range, IQR) in mutational tolerance across homologs at each position (Fig. 1b, c). The considerable spread in mutational tolerance highlights that positions allowing few mutations in one homolog may show substantial mutability in other homologs. A median IQR of 0.87 means that half the positions have a >1.8-fold ($2^{0.87} = 1.83$) difference in effective number of tolerated amino acids between homologs. IQR can be as high as 2.51, indicating a 5.7-fold difference in effective amino-acid number. Furthermore, the difference in minimum and maximum mutational tolerance across positions has a median of 2.69 and a max of 3.79 (Fig. 1c), meaning half of the positions in the protein have a 6.5–13.8 fold difference between the most and the least mutationally tolerant sequences. This strong mutational heterogeneity of equivalent

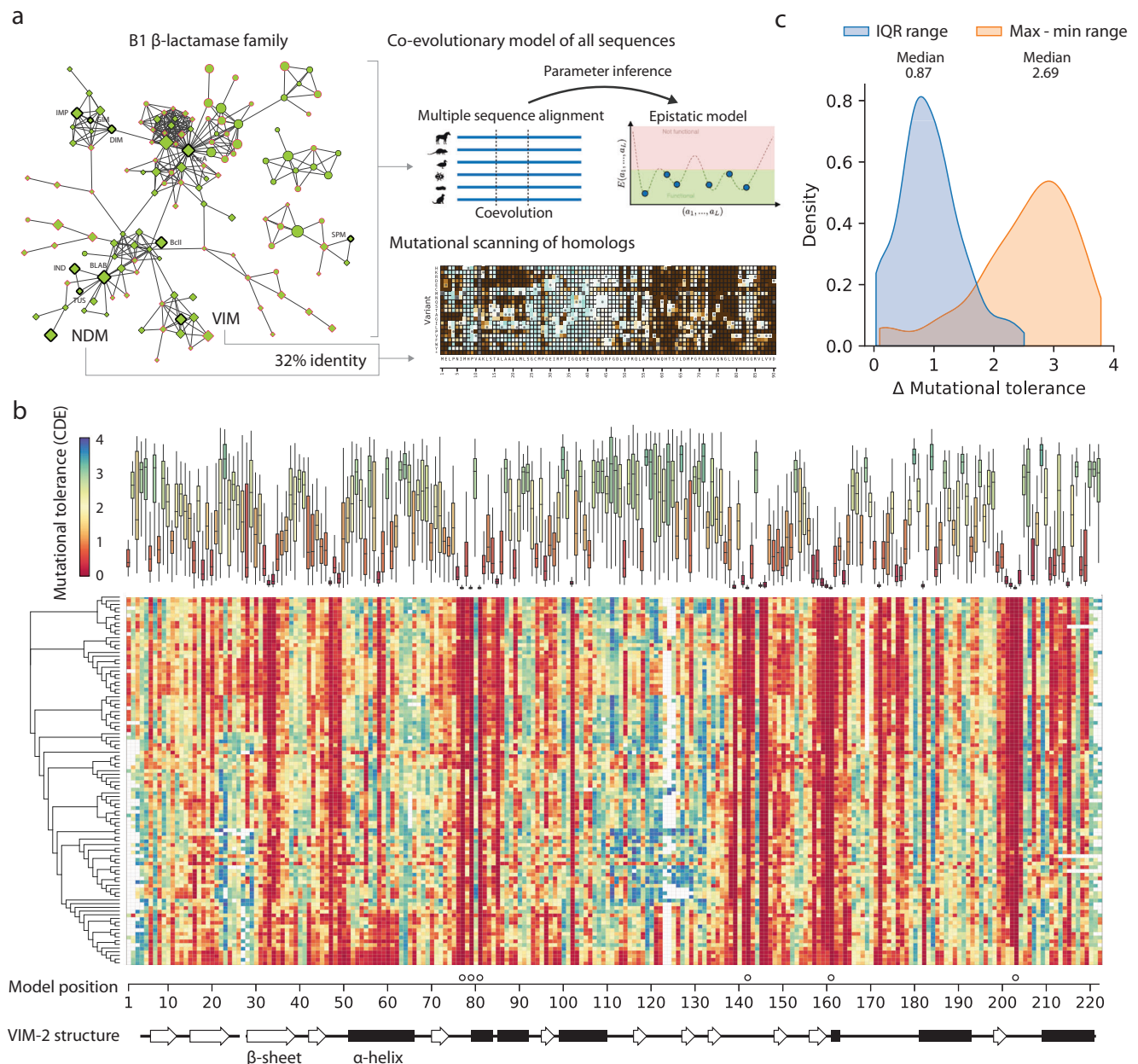


Fig. 1 | Family-wide residue mutability by direct coupling analysis (DCA).

a Schematic of computational and experimental workflow. The B1 β -lactamase enzyme family is isolated through sequence space exploration via a sequence similarity network. The entire set of sequences is used to generate a co-evolutionary model via DCA. Two highly diversified homologs within the family are selected for DMS to generate a large mutational dataset. **b** Each square in the heat map is colored by the predicted mutational tolerance (measured as context-dependent entropy, CDE) for each position of the 100 aligned homologs. Blank cells represent alignment gaps. The distribution of CDEs by position is presented as

box plots (N =number of aligned homologs at position, exact count in source data) on top: 0–100% as whiskers, 25–75% (IQR) as bars. The bars are colored by the median value, with the same color scale as the heatmap. The secondary structure of a representative (VIM-2, PDB ID: [SYD7](#)), as well as the active site residues (circles), are depicted under the heat map; The maximum likelihood phylogenetic tree of the 100 homologs is shown on the left of the heat map; arrows represent β -sheets, bars represent α -helices. **c** Distributions of the per position spread in mutational tolerance across homologs, measured as the distance (in bits) of the IQR or max-min mutational tolerance. Source data are provided as a Source Data file.

positions across homologs is a hallmark of epistasis, and is captured by the coevolutionary couplings in the DCA model.

Heterogeneity in mutational behavior is supported by DMS data of NDM-1 and VIM-2

To experimentally observe differential mutational effects between homologs, we measured all single-mutational effects on NDM-1 and VIM-2 (homologs of ~30% sequence identity) by DMS. We previously published the DMS dataset of VIM-2⁵⁰, and we performed DMS on NDM-1 in an identical manner (Fig. 2a). Briefly, all single amino acid

mutants generated for NDM-1 were selected with three different antibiotics: ampicillin (AMP), cefotaxime (CTX), and meropenem (MEM). Plasmid DNA isolated after selection was deep sequenced, and we calculated the fitness conferred by each mutant relative to wtNDM-1 in Eq. (1) (Fig. 2a, methods, Supplementary Data 3). The two replicates show good correlation (R^2 of 0.77–0.95) (Fig. 2b, Supplementary Fig. 2). Overall, the three antibiotics behave similarly, and thus, for the best comparison, we use datasets between NDM-1 and VIM-2 with similar selection pressures in AMP⁵⁰ (Supplementary Fig. 3).

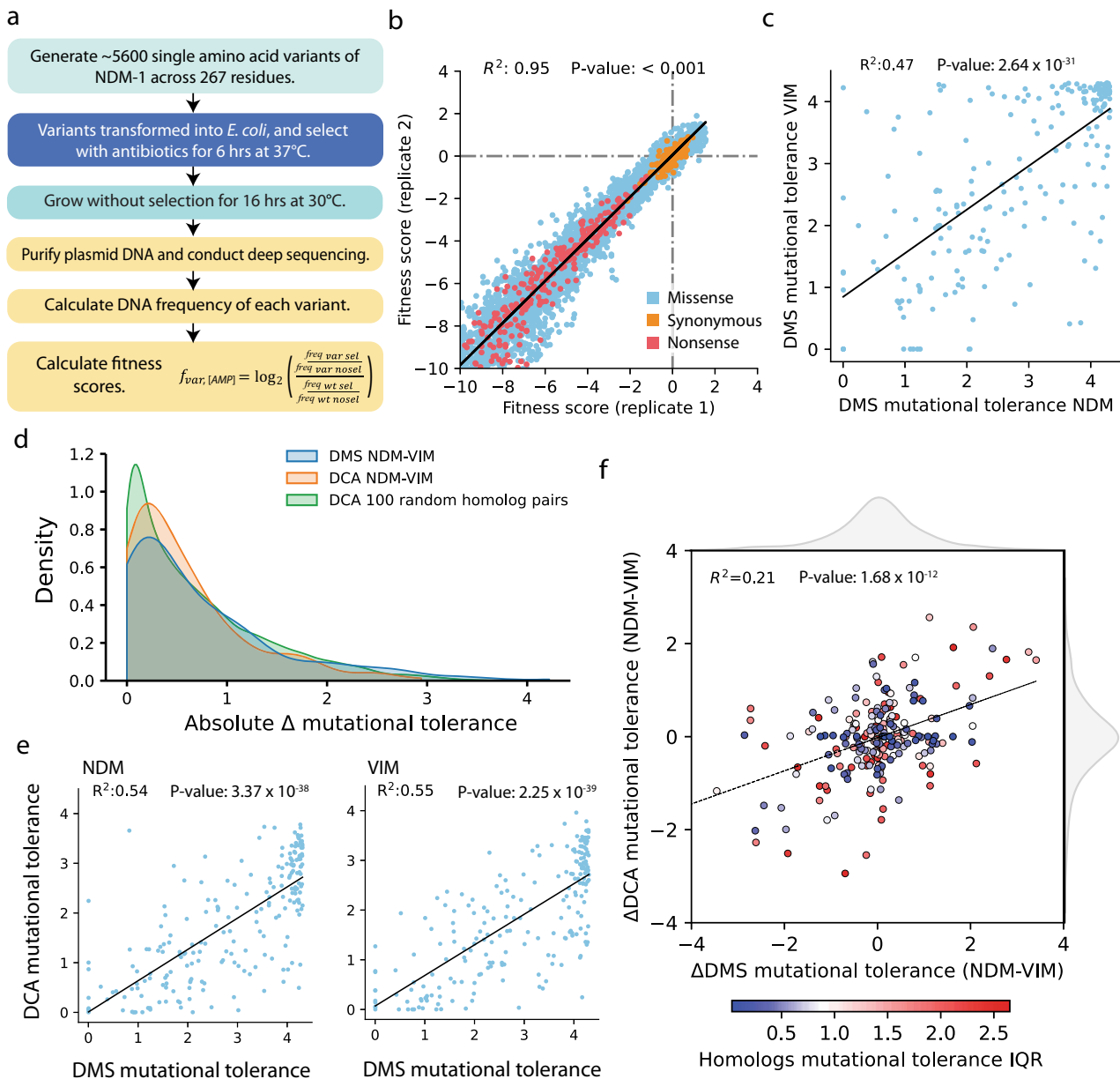


Fig. 2 | Overview of DMS for NDM-1 and structural similarity to VIM-2.

a Workflow for DMS of NDM-1. **b** Correlation between replicates of the NDM-1 library selected at 256 μ M AMP. The R^2 and P -value of a linear regression are shown at the top. **c** Comparisons of mutational tolerance at each aligned position for the NDM-1 and VIM-2 experiments. The R^2 and line of best fit for a linear regression are shown. **d** Distribution of differences in mutational tolerance at the same aligned position in DMS or DCA between NDM-1 and VIM-2, or the difference in DCA between 100 random pairs of homologs. **e** Comparison of mutational

tolerance between DMS and DCA for NDM-1 and VIM-2. The R^2 and line of best fit for a linear regression are shown. **f** Comparison of the mutational tolerance difference between NDM-1 and VIM-2 at each position between DMS and DCA. The data is colored by the IQR of mutational tolerance across the 102 homologs (100 + NDM-1 and VIM-2), with the colors scaled to the distribution (median is the center). The R^2 and line of best fit for a linear regression are shown. Source data are provided as a Source Data file.

Similar to DCA, the DMS mutational tolerance of each position is calculated as the Shannon entropy of the normalized fitness of all measured mutants (methods). Mutational behaviors in equivalent protein positions are varied between the two homologs, and certain positions can accept only the WT residue in one homolog (entropy near 0) and almost all amino acids in the other homolog (entropy near 4) (Fig. 2c). Comparing DMS and DCA, we find strikingly similar distributions in the entropy differences between NDM-1 and VIM-2 (Fig. 2d). The median difference between homologs across positions is 0.45 in DMS (0.39 in DCA), meaning half of the positions have a > 1.37 -fold difference in entropies, up to a maximum of 4.22 (18.7-fold). As a baseline, we also computed the CDE difference between 100 random

pairs of homologs (Fig. 2d). We again found a similar distribution, suggesting a common statistical trend where certain positions exhibit similar tolerance (such as conserved sites) while others can greatly vary in mutation tolerance.

We check agreement of DMS and DCA data on each specific position within either NDM-1 or VIM-2 (Fig. 2e). There is a significant correlation ($R^2 \sim 0.55$) between the DMS and DCA values for each of NDM-1 and VIM-2, showing that DCA captures sequence-specific trends to a good degree. We also compare the DMS mutational tolerance to the context independent entropy (CIE), the entropy calculated using just single position information in the MSA without coupling to other positions (Supplementary Fig. 4), and find the DCA mutational

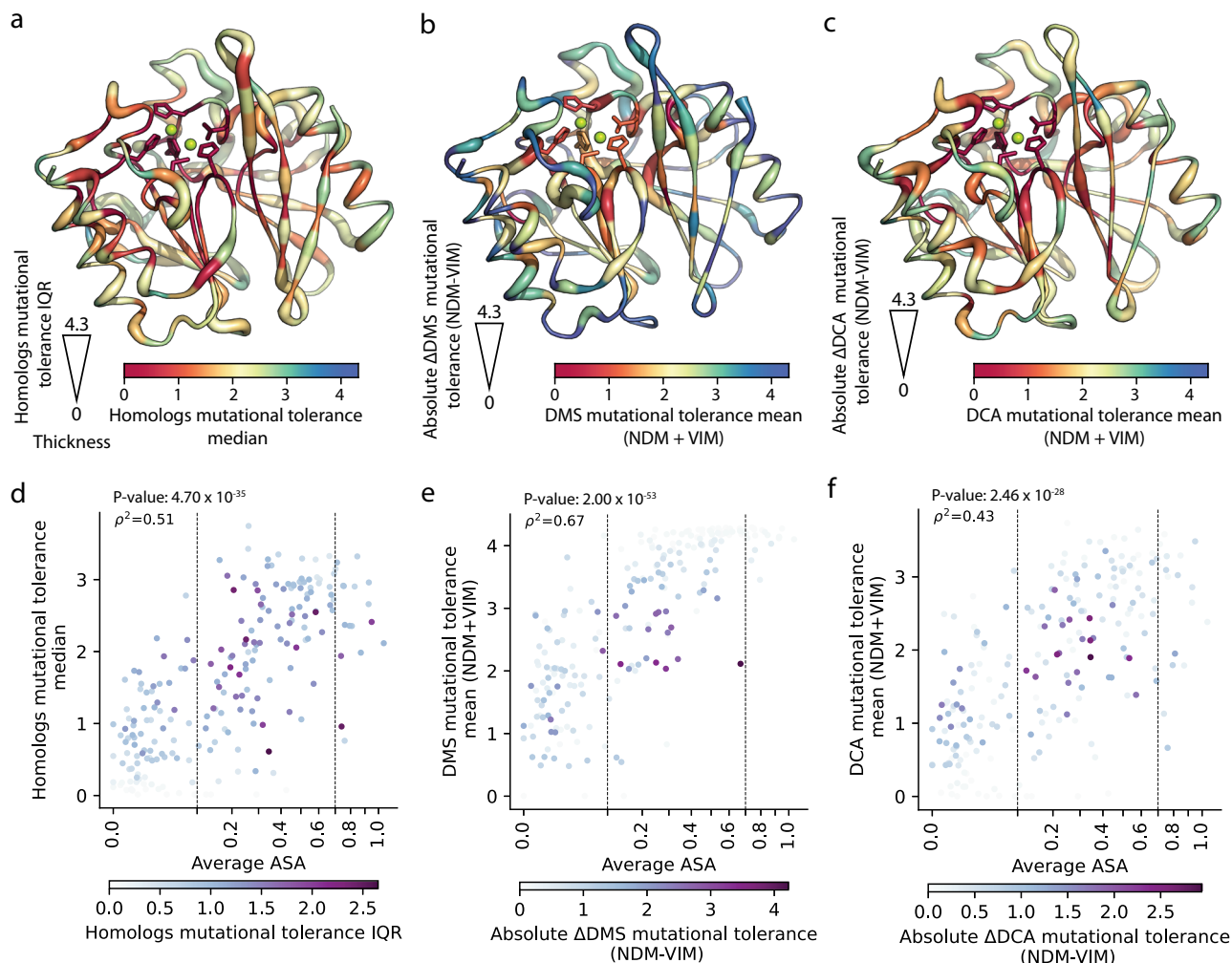


Fig. 3 | Structural basis of tolerance classifications. **a** All homologs mutational tolerance data overlaid on the crystal structure of VIM-2 (5YD7), with the thickness of the backbone representing the IQR in the 102 homologs, and colored by the median. **b** DMS mutational tolerance data overlaid on the crystal structure of VIM-2, with the thickness of the backbone representing the absolute difference between NDM-1 and VIM-2, and colored by the average mutational tolerance. **c** DCA mutational tolerance of VIM-2 and NDM-1 overlaid on the crystal structure of VIM-2, with the thickness of the backbone representing the absolute difference, and colored by the average. **d** Scatter plot of the median mutational tolerance values of 102

homologs versus the average ASA of VIM-2 and NDM-1, with positions colored by the mutational tolerance IQR. **e** Scatter plot of the average DMS mutational tolerance of NDM-1 and VIM-2 DCA CDE versus the average ASA, with the positions colored by the difference in mutational tolerance between NDM-1 and VIM-2.

f Same as panel **e** but for the DCA predictions: scatter plot of the average mutational tolerance of VIM-2 and NDM-1 versus their average ASA, colored by the mutational tolerance differences. In **d–f**, the ρ^2 and p -value of a Spearman correlation between x and y variables are shown. Source data are provided as a Source Data file.

tolerance (which uses sequence context) to perform better (R^2 for CIE ranges from 0.45 to 0.48). It is important to note, using the sequence context not only improves prediction of mutational tolerance over CIE, but also allows for quantifying sequence specific behaviors that would not be possible with CIE which ignores sequence context. However, DCA predicts fewer variable sites (entropy >3.5) compared to DMS, shown by the points below the diagonal in Fig. 2e. Indeed, it has been observed that DCA predicts a Gaussian-like distribution of mutational effects⁴⁸, and tends to underestimate the number of neutral mutations which are often one peak in a bimodal distribution^{3,5,7}. The discrepancy may arise since the DMS data only captures a single selection antibiotic, while DCA reflects broader evolutionary constraints. We then find that DCA can also capture the specific differences between homologs in DMS (Fig. 2f); the abundance of correctly predicted small differences weakens the correlation. When the variability (IQR) in DCA mutational tolerance across all 102 homologs (100 + NDM-1 and VIM-2) is overlaid on the plot, we see that positions with the lowest spread in all homologs tend to show lower differences between NDM-1 and VIM-2 in both DMS and DCA, as expected (Fig. 2f). Meanwhile, positions

with large differences across all homologs are spread throughout. For positions with large differences in DCA or DMS, the varied behaviors of NDM-1 and VIM-2 match the global variation. Interestingly, there are positions with small differences between NDM-1 and VIM-2, but having high variability across all homologs, underscoring the fact that mutational behaviors can vary strongly between different homologs. Hence, the combination of methods can reveal and reinforce patterns that would not be obvious from just a single approach.

Structural basis of mutational tolerance and incompatibilities

We investigate the relationship between mutational heterogeneity and structure of the B1 family, using crystal structures of NDM-1 (PDB ID:3SPU) and VIM-2 (5YD7) as representatives. The emphasis placed on structure is based on previous linear modeling of mutational effects on VIM-2, where structural features explained the largest amount of variation in the data, while other factors such as residue size or charge had little explanatory power for this system³⁰. We superimpose the global variability (IQR) and median mutational tolerance of the 102 homologs on the structure (Fig. 3a). We produce equivalent figures using the

local DMS (Fig. 3b) and DCA (Fig. 3c) data of NDM-1 and VIM-2, using variability (difference) and mean mutational tolerances. We see similar tendencies in the three figures. Buried regions in the protein core, including the active-site metal-binding residues, tend to have both low mutational tolerance and spread; likely being critical to folding or activity. The most exposed positions generally have high entropy and very low spread in both DMS and DCA, as these positions are highly mutationally tolerant. Finally, some buried residues have higher entropy in DMS than in DCA, which could be due to different residue and spatial arrangements specific to the two homologs compared to the global distribution of the family.

To quantify the role of structure, we use the average accessible surface area (ASA) of NDM-1 and VIM-2. Absolute ASA from the NDM-1 and VIM-2 structures measured in PyMol are normalized to an ASA reference set⁵¹ before averaging. There is significant correlation between average ASA and the site-specific mutability, evident in both experimental data and model predictions (Fig. 3d–f). Correlation between ASA and mutability is commonly observed^{3,7,52–54}, likely as internally situated residues have higher prevalence of structural interactions, amplifying the potential for mutations with adverse effects. Furthermore, we used the ASA as a structural variable to distinguish three classes of residues: very buried ($ASA < 0.1$), partially exposed ($0.1 \leq ASA < 0.7$), and very exposed ($ASA \geq 0.7$) (Supplementary Fig. 5). Very buried residues are more mutationally constrained, with low entropy and low spread, where some positions also exhibit large mutational variability between NDM-1 and VIM-2 (Fig. 3e, f). Meanwhile, the very exposed positions typically display high mutational entropy and low spread, displaying a near absence of mutational heterogeneity in VIM-2 and NDM-1. Most importantly, we find an intermediate region of partially exposed residues showing some residues with very high spread in mutational tolerance in both DMS and DCA (Fig. 3d–f). In terms of overall distribution, the partially exposed region tends to have greater mutational heterogeneity in DCA compared to the buried region (1-sided Brunner–Munzel test, p -value < 0.05), but the buried region tends to have greater mutational heterogeneity in DMS compared to the partially exposed region (Supplementary Fig. 5a, b). Again, this appears to be a difference between global and homolog specific effects, as the differences of DMS mutational tolerance between NDM-1 and VIM-2 specifically is much lower for many positions than with the 100 other homologs (also reflected in the DCA distributions comparing just NDM-1 and VIM-2, Supplementary Fig. 5c). Regardless, the positions with the strongest mutational heterogeneity are always in the partially buried regions.

These observations are possibly due to more intramolecular interactions around buried residues, whereby a rich intramolecular network reduces the mutability of residues, but also leads to homolog-specific differences of such constraints. However, it is the partially exposed region that exhibits the largest variability in behaviors for all datasets. Possibly, this range corresponds to positions with more mutational freedom than fully buried positions, while still forming interactions with other residues. The outcome of a mutation is therefore strongly dependent on the sequence context, i.e., homolog-specific. The large mutational variability of the partially exposed region is common to all analyses (Fig. 3d–f), affirming the dependence between epistatic networks and the structure.

Probing the influence of epistatic networks on specific mutations

To characterize the epistatic networks and reveal the interdependencies between residues, we compare the effects of individual variants in the DMS of NDM-1 and VIM-2. Variants can be compared directly for positions where VIM-2 and NDM-1 have the same WT amino acid (81 positions), referred to as ‘shared’ positions. However, at ‘differing’ positions with different WT residues (136 positions), the difference in mutational starting point prevents a direct comparison;

unlike mutational tolerance, which already accounts for all mutational states (including WT), individual mutational effects relative to the WT need to consider the specific WT residue. We hence analyze mutations in ‘shared’ and ‘differing’ positions using different schemes (Fig. 4a, methods). At differing positions, we first check the effect of swapping WT residues at each position between NDM-1 and VIM-2 (WT reversions, 132 comparisons and 4 reversions not observed), and if the swap is neutral in both homologs (classified as ‘compatible’, total of 50 positions) we treat them like the ‘shared’ positions (131 total including ‘compatible’ positions) (Supplementary Data 4). Between NDM-1 and VIM-2, individual mutational effects are very different (Fig. 4b). To determine if a mutation is epistatic between backgrounds, we conduct z-tests of the difference in fitness (NDM minus VIM) against a null distribution with the same standard deviation as the synonymous variants distribution of NDM-1 (greater standard deviation of the two backgrounds). We find ~28% of individual mutations in the shared positions are epistatic and that differences are not strongly biased in sign or effect size toward a single homolog (Fig. 4c). We compute the fraction of epistatic mutations per position as $N_{\text{mut epistatic}}/N_{\text{total observed}}$, revealing widespread epistasis across the entire structure (Fig. 4d). The median fraction of epistatic mutations at each position is 0.21, meaning that half of the positions have significant epistasis in at least ~4 mutations (0.21×19 missense a.a.). Notably, strongly epistatic positions are enriched in partially exposed positions (Fig. 4e), as previously noted (Fig. 3d–f). In the case of individual mutations, the partially exposed distribution tends to have the higher epistatic effects relative to both buried or exposed regions (1-sided Brunner–Munzel test, p -value < 0.05 , Supplementary Fig. 6a), in addition to containing some of the highest epistatic effects.

For differing positions, swapping WT residues between NDM-1 and VIM-2 is deleterious at 82 positions. We classify the positions as ‘1-WT entrenched’ if the swap is incompatible in one background (43 positions) and ‘2-WT entrenched’ if the mutation is incompatible in both backgrounds (39 positions) (Fig. 4f). We use the term ‘entrenchment’ to describe the inability to access a certain amino acid in the presence of other mutations that have already taken place, when that same amino acid is viable in another context; i.e., a residue becomes entrenched after accruing a certain amount of mutations relative to the other background. We note that this is entrenchment based strictly on a hypothetical mutational path between NDM-1 and VIM-2, and is an aggregate of all final mutational interactions regardless of the exact path taken. Again, we see pervasive epistasis, where 42–49% of WT-swapping mutations (65/132 in NDM-1, 56/132 in VIM-2) lead to a significant loss in fitness. Structurally, entrenched positions are scattered across the whole protein (Fig. 4g) with a subtle bias for 1-WT entrenched positions to be less buried than 2-WT entrenched positions (Supplementary Fig. 6b). To further explore the role and meaning of entrenched positions, we use DCA to calculate ΔE of swapping entrenched WT residues between NDM-1 and VIM-2 (Fig. 4h). Similar to DMS, mutations concentrate around the neutral center, or populate the lower half-plane defined by the diagonal. As expected, almost no points populate the upper half-plane where the swap of both WT residues would result in higher fitness. Without epistasis, the mutations $A \rightarrow B$ and $B \rightarrow A$ in arbitrary backgrounds should have opposite signs, exhibiting a perfect anticorrelation along the diagonal in Fig. 4h. The distance away from the diagonal is the difference in mutational effect in the two sequence contexts, and thus a rigorous signature of epistasis. We verify coherence between DCA and DMS by checking the fraction of mutations from each entrenchment class at different distances from the diagonal (Fig. 4i), and we do see 2-WT entrenched mutations enriched further from the diagonal, while those closest to the diagonal are mostly compatible.

Once again, the model and the experiment complement each other, both confirming epistatic interactions to be pervasive, and associated with structure. Moreover, DCA proves to be quite accurate in predicting epistasis of specific mutants, as all the most distant points

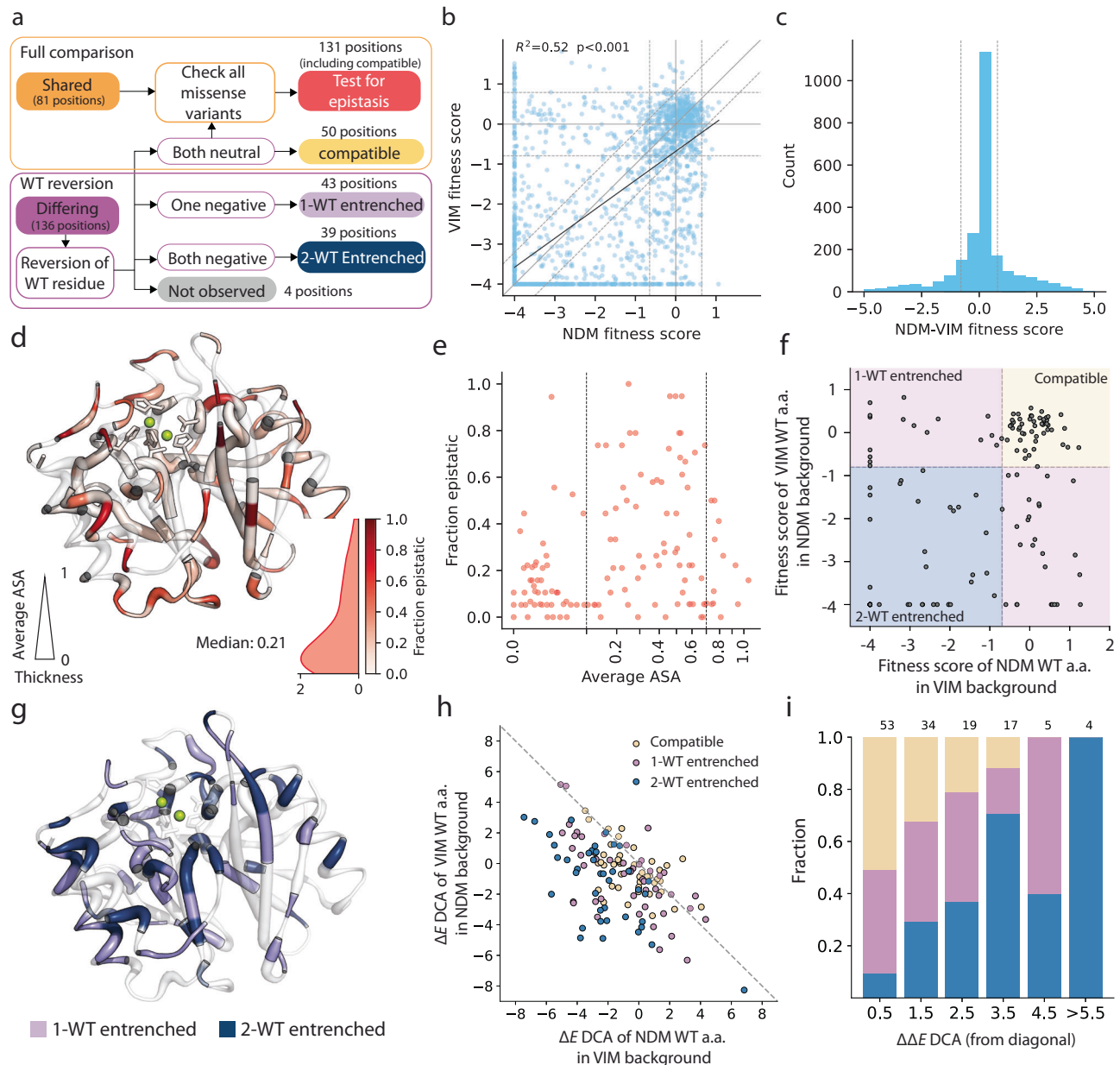


Fig. 4 | Residue level epistasis between NDM-1 and VIM-2. **a** Flowchart of epistasis classification. **b** Correlation of DMS data between NDM-1 and VIM-2 at shared and compatible positions. Dashed lines highlight range of neutral fitness effects ($1.96 \times$ standard deviation (SD) of synonymous variants distribution of respective homolog, diagonal uses y-range, centered on 0 (axes) or the 1:1 line (diagonal)). The fit line (linear correlation) is shown in black, with R^2 and p -value displayed. **c** Distribution of fitness effect differences between NDM-1 and VIM-2 at shared and compatible positions. Dashed lines represents the range of neutral fitness around 0 ($1.96 \times$ SD of synonymous variants for NDM-1). **d** Fraction of epistatic mutations ($N_{\text{mut epistatic}}/N_{\text{total observed}}$) by position overlaid on the VIM-2 structure (shared and compatible positions, others transparent), colored according to the lower right (with distribution). Thickness of the structure corresponds to the average ASA of the NDM-1 and VIM-2 crystal structures. **e** Plot of the fraction of epistatic mutations at each position versus the ASA. Only positions highlighted in panel **d** are included.

f Scatter plot of fitness effects for reverting VIM-2 WT to NDM-1 WT (x-axis) and NDM-1 WT to VIM-2 WT (y-axis) for equivalent positions. The vertical dashed line indicates the lower bound of the region of neutral effects for VIM-2 based on the synonymous variant distribution, and the horizontal dashed line shows the lower bound of the region of neutral effects for NDM-1. Quadrants with different behavioral classes are colored as in **a**. **g** Differing WT positions between NDM-1 and VIM-2, colored by entrenchment class. The structure thickness corresponds to average ASA and regions outside the classification are transparent. **h** Scatter plot of ΔE for reverting VIM-2 WT to NDM-1 WT (x-axis) and NDM-1 WT to VIM-2 WT (y-axis) for equivalent positions, colored as in **a**. The dashed line represents the expected behavior without epistasis. **i** Bar plot showing the relative fraction of points in each epistatic class at various distances from the diagonal of **h**. The total number of points in each distance bin is shown on top. Source data are provided as a Source Data file.

from the diagonal are either 1- or 2-WT entrenched according to the experiments (Fig. 4i).

Epistatic positions highlight complexity of interaction networks

Our data provides the opportunity to dissect specific epistatic interactions in the homologs, as 2-WT entrenched positions may indicate

the residues participate in specific interactions. We sought these interactions within the proteins through pairs of positions that are in close proximity, and are both 2-WT entrenched (Supplementary Fig. 7a). One example of shape complementarity can be found between NDM-1 positions 125 and 249 (VIM-2 positions 119 and 239) (Fig. 5a), under the active-site zinc ions. In VIM-2, the Arg at position 125 is

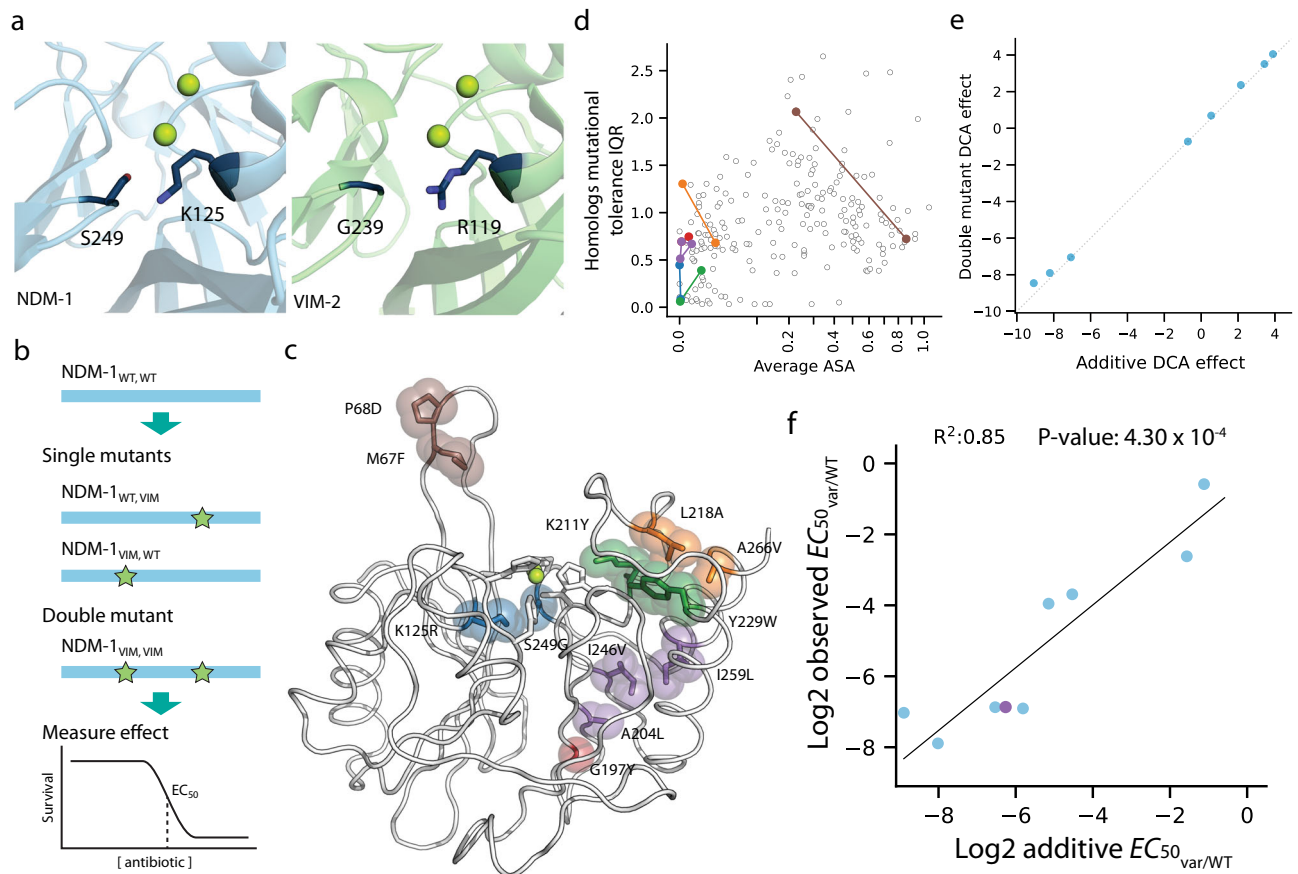


Fig. 5 | Testing interactions of entrenched positions in NDM-1. **a** Example of potentially interacting entrenched WT positions in the crystal structures of NDM-1 and VIM-2. **b** Experimental scheme for testing single or combined mutational effects in the NDM-1 background. **c** Entrenched WT positions that were chosen for testing of epistatic interactions. Positions with the same color are mutated together to test for compensation of entrenchment; A204L overlaps 2 sets and is also tested with G192Y (red). **d** Plot of IQR in mutational tolerance across 102 homologs and the average ASA of NDM-1 and VIM-2 structures, with the tested positions highlighted.

Tested combinations are shown as lines. **e** Scatterplot of DCA energy change of all selected double mutants, with the expected additive single mutant effects in the x-axis, and the observed double mutant effects in the y-axis. Dashed line indicates a 1:1 correlation. **f** Scatterplot of all tested double (1 triple, purple) mutants, with the expected additive single mutant effects in the x-axis, and the observed double mutant effects in the y-axis. Effects are calculated as fold change relative to wtNDM-1. The line of best fit for a linear correlation is shown in black, with the R^2 and p -value displayed. Source data are provided as a Source Data file.

paired with Gly at position 249. In contrast, NDM bears a smaller but still positively charged Lys at position 125, which is now neighbored by a larger Ser at position 249. To test if these positions possess significant interactions, we measured the phenotype (EC_{50} against AMP) of the single and double mutants of each position pair in the NDM-1 background, mutated to the VIM-2 WT residues (Fig. 4c). As VIM-2's WT residues are deleterious in NDM-1 as single mutants, we expect that mutating both positions may lead to compensation, generating a less negative effect. We also test pairings in the L3 active site loop (NDM-1 positions 67 + 68), L10 active site loop (218 + 266, 211 + 229) and some buried positions beneath the L10 loop (197 + 204, and a triplet of 204 + 246 + 259) (Fig. 5c); the triplet was tested in pairs and in triplet. Combined, our selection of positions covers a variety of positions, with different biochemical properties (size, polarity, charge), different solvent accessibilities and different variability in mutational behaviors across the whole family (Fig. 5c, d).

The single mutant effects confirm the entrenchment observed from the DMS fitness scores, as all selected WT residues in VIM-2 are deleterious in the NDM-1 background, evenly spanning a wide range of deleterious effects (Supplementary Fig. 7b, Supplementary Data 5). The EC_{50} and the fitness score form a sigmoidal relationship, consistent with previous observations⁵. However, when we tested the mutants in combination, we did not observe any significant compensation (Fig. 5f). The log-additive effects of the single mutants (null

model for no epistasis) show a distinctly linear correlation with the observed double mutant effects (R^2 of 0.85). It appears that entrenched positions cannot be swapped simply by mutating other nearby entrenched residues. This scenario is confirmed by DCA, where larger epistatic effects can only arise as a cumulation of small contributions from many epistatically coupled positions (Fig. 5e). Previous experiments have also shown that direct compensation by nearby residues is expected to be rare¹³, and compensation often requires more than two mutational steps⁵⁵. The double mutation effects are therefore additive in DCA (Fig. 5e) and in the experiment (Fig. 5f). Notably, the positions are not globally independent in their effect, as these mutations are fixed in VIM-2 and the deleterious effects have been compensated for during evolution. Thus, these experiments suggest the intramolecular network of residue interactions to be complex, and even entrenched residues are not themselves sufficient to dictate the epistatic networks.

Discussion

By combining the computational approach of DCA and the experimental approach of DMS, we have gained a global picture of the epistatic tendencies between the homologs within a protein family. In the B1 β -lactamase family, we find a prevalence of heterogeneity in mutational effects that is jointly revealed by the two methods. At a global level across all homologs in DCA, over half of the positions can exhibit a >6 fold difference in mutational tolerance depending on the

sequence background. When examined specifically in the NDM-1 and VIM-2 contexts, half of positions with shared WT residues have >4 mutants that are epistatic in DMS. In particular, the complementarity of the two approaches has enabled better understanding than either approach alone. Notably, the use of DCA greatly expands the number of proteins we can analyze. The global level view provided by DCA for 102 homologs reveals the full spread in mutational behaviors, which would have been obscured for positions that behave similarly for the 2 specific homologs tested experimentally. Although DCA is purely computational in nature, the ability of DCA to systematically capture the effect of epistasis on mutations in proteins has been thoroughly demonstrated in a variety of applications. This includes the ability to predict DMS data⁴⁷, *E. coli* strain polymorphisms^{48,56}, modeling of evolution experiments⁵⁷ and generating highly diverged but functional artificial sequences up to 70–80% divergence²⁶. Even if there may be noise in the DCA predictions, we believe our use of the method mitigates the effects of noise on our findings as we largely use DCA to examine broad level trends rather than exact differences between specific data points. Regardless, we have shown that trends from both DCA and DMS tend to agree quite well (e.g., Fig. 3), even when directly considering just differences between NDM-1 and VIM-2 (Figs. 3c, f, 4h, i). In turn, while both approaches generally agree on the mutational behaviors of each position, the experimental results can highlight background specific behavior, such as higher mutational tolerance in NDM-1 or VIM-2 than expected by the global DCA model (Fig. 2e).

To extend upon previous DMS studies involving multiple homologs^{29–32,34,35}, we also performed deeper examination of the mechanisms behind the observed incompatibilities. The ASA of a protein position shows trends with various facets of epistasis, as global spread is more constrained in behavior at extremely low or high ASA, while residues with the strongest epistatic behavior appear at intermediate ASA. These observations are likely underpinned by differential intramolecular networks in different homologs, as a result of gradually co-evolved epistatic networks over the course of evolution. However, directly replacing just two or three residues is not enough to compensate for evolutionarily entrenched residues, suggesting that a much more complex network of interactions is at play in line with previous findings^{13,55,58}, and as predicted by the DCA model in the current study.

We suspect that many of the observed behaviors are likely not limited to NDM-1 and VIM-2, or the B1 family. The trends with regards to structure in both DCA and DMS, which are typical of those observed in other systems^{3,59–62}, suggest that the behaviors we observe can be explained through general mechanisms such as secondary structure formation and structural packing. However, more evidence from other systems would be required to distinguish the global trends from system-specific trends. Overall, we provide a global view of epistatic networks at the protein family level that complements more detailed examinations of epistasis in specific residues^{32,63}.

Methods

Sequence collection for the B1 MBL family

To conduct DCA, we first collect sequence data of the B1 MBL family. In an effort to comprehensively isolate all B1 MBLs, we used a broad sweep approach through a sequence similarity network (SSN) using data from genomic (UniProt) and metagenomic (JGI) databases. Initially, an SSN was constructed using the EFI-EST tool using the InterPro ID IPR001279 (MBL domain superfamily) as the input. The network was analyzed at different alignment score cut-offs in Cytoscape (v3.8) to find a cut-off where all clinically isolated B1 enzymes fell within the same isolated cluster (~1800 sequences at EFI alignment score cut-off of 25). The sequences that were shorter than 220aa were removed, and the remainder (~1300)

were clustered by CD-hit (v4.8.1) to 60% identity to reduce redundancy (187 sequences final) then used to generate a Hidden Markov Model (HMM) of the B1 family using HMMER (v3.3.2). The HMM was used to search for B1 sequences in the Joint Genomics Institute Integrated Microbial Genomes & Microbiomes (JGI IMG/M) database resulting in ~2.5Mil sequences from JGI, with 1,859,503 non-redundant sequences determined by CD-hit clustering at 100%.

To construct the final SSN, all sequences of the MBL domain superfamily were downloaded from UniProt using the InterPro IPR001279 definition (434,623 sequences, accessed 2019). The UniProt and JGI data were combined, sequences with ambiguous characters were removed, and the length was restricted to between 100 and 1500aa. CD-hit was used to successively cluster the sequences to 100%, 90%, 70% and finally 50% identity to reduce redundancy, resulting in 86,947 representative sequences for the entire MBL superfamily. The SSN was generated by performing ‘all by all’ BLASTP (BLAST+ v2.9.0) with an e-value cutoff of 1e–5 and 1000 maximum hits per sequence, giving a raw network of ~90Mil pairwise bitscore values (edges) between all sequences (nodes). The network was processed using an inhouse SSN analysis pipeline, MetaSSN⁶⁴ (<https://github.com/johnchen93/MetaSSN>) to identify the lowest bitscore cutoff at which all clinically isolated B1 sequences break off into an isolated cluster (10,252 sequences). Finally, to identify sequences that are most likely to be active B1 sequences, the dataset was filtered by length to between 200–350aa (6,828) and used to build a multiple sequence alignment (MSA) using Clustal Omega (v1.2.2) on default settings. The MSA was manually curated to exclude any obviously mis-aligned sequences (e.g., shifting of entire sequence, large gaps in generally conserved columns), resulting in 6308 curated sequences. To ensure the sequences were likely to have B1-like function, only sequences aligned with the conserved B1 active-site metal binding residues (H116/H118/H196 and D120/C221/H263, BBL numbering) were kept, resulting in a final dataset of 5035 B1 family sequences, with a roughly 50/50 split between sequences from UniProt and JGI. Finally, all B1 sequences in this study were joined with all B1 sequences from the BLDB (481), resulting in 5516 sequences total (4046 non-redundant at CD-hit 100%).

Direct coupling analysis of the B1 family

To conduct DCA, we build a multiple sequence alignment of all B1 MBL sequences. To generate an alignment of conserved structure, crystal structures of 14 homologs were aligned using mTM-align (<https://yanglab.nankai.edu.cn/mTM-align/>)⁶⁵. The homologs (PDB IDs) are: NDM-1 (3spu), VIM-2 (5yd7), DIM-1 (4wd6), ECV-1 (6t5k), FIM-1 (6v3q), GIM-1 (2ynt), IMP-1(4uam), IND-7 (3l6n), MYO-1 (6t5l), TMB-1 (5mmd), VMB-1 (6jv4), bc-II (1bc2), blaB (1m2x) and cfiA (1znb). Positions outside the structurally conserved regions were excluded. An HMM sequence profile was trained on this curated dataset by using HMMER via the *hmmbuild* command. We used a *-symfrac* value of 0.3 to control the maximum number of gaps in each alignment column. All B1 sequences previously collected were then aligned to this profile via the *hmmsearch* command to produce the MSA for training the DCA model. After converting the alignment to FASTA format, all sequences exhibiting more than 10% gaps were removed from the alignment. Additionally, flank columns exhibiting more than 75% gaps were eliminated. The resulting alignment contained 222 positions (Supplementary Data 6).

To select 100 homologs (in addition to NDM-1 and VIM-2) for further analysis by DCA, all sequences were clustered using CD-hit at 80% sequence identity. Then 100 clusters were selected at random and the representative sequences were used as the target homologs. The resulting sequences were highly diverse (~30% sequence identity to each other). A final refinement to the MSA used to train the DCA model was achieved after removing all sequences that exhibited

more than 80% sequence identity to NDM-1, VIM-2, or the 100 homologs thereby ensuring the alignment was not biased toward the sequences used for further analysis. The resulting MSA had 3003 sequences. We then inferred on this alignment a DCA model by using standard settings of the adabmDCA package corresponding to Persistent Contrastive Divergence using 40 MC sweeps at each iteration (PCD-40)²⁰.

Library generation and deep mutational scanning of NDM-1

The procedure for library generation and DMS on NDM-1 were conducted in an identical manner to VIM-2. The wtNDM-1 sequence is encoded on an inhouse pIDR5.1 plasmid, expressed under a constitutive AmpR promoter (Supplementary Data 7). To generate all single amino acid mutants, we used a PCR based method (restriction free cloning⁶⁶) to introduce a degenerate 'NNN' sequence at each codon in the coding sequence in separate reactions. After each single position was mutated, the library was combined into 7 groups of 39 consecutive positions each, forming 117nt long mutated regions that can be fully sequenced by paired end Illumina NextSeq reads. All primers were purchased from IDT (Supplementary Data 8).

The NDM libraries are then transformed into *Escherichia coli* (*E. coli* 10 G, Lucigen) and stored as glycerol stocks, with the number of colony forming units after transformation measured to be >100,000 to ensure complete coverage of each group. To perform selection experiments, we inoculate the glycerol stocks into fresh LB (Thermo Fisher) and grow the cultures shaking overnight at 30 °C for 16 h. The cultures are diluted into fresh LB to an inoculum of 1.5×10^6 cells/mL (targeting a 1:1000 dilution of a culture with OD₆₀₀ of 1.5) and grown shaking at 37 °C for 2 hrs. Selection pressure is introduced by mixing 960 µL of cell culture with 40 µL of a 25× stock of each antibiotic suspended in LB, for a final culture volume of 1 mL. We tested selection at 32, 256 and 512 µg/mL AMP (Thermo Fisher), 2, 16, 32 µg/mL CTX (Thermo Fisher), and 0.063, 0.25 and 0.5 µg/mL MEM (Sigma Aldrich). We also grew a sample of the library without selection. The cultures (with and without selection) are grown while shaking at 37 °C for 6 hrs, then removed from selection via centrifugation and resuspension in 1 mL of fresh LB, repeated 3 times. The post selection culture is grown, shaking overnight at 30 °C for 16 hrs and the plasmid DNA is purified using a QiaPrep 96 column DNA purification kit (Qiagen). This procedure was conducted twice on different days to produce two separate replicates.

To deep sequence the selected library, we used primers targeting unmutated regions that directly flank the mutated region of each of the 7 library groups to amplify the DNA and to attach Nextera adaptors to the amplicons. The amplicons undergo a second PCR to attach the Illumina sequencing indices and flowcell binding sites. All tested samples were sequenced in the same Illumina NextSeq 550 run with fully overlapping paired end reads. We also included control samples of amplicons extracted from just wtNDM-1. Raw sequencing data is available on the NCBI Sequence Read Archive under BioProject PRJNA974578.

After deep sequencing, the forward and reverse reads are merged together using our previously published pipeline, DMS-FastQ-processing⁶⁷ (<https://github.com/johnchen93/DMS-FastQ-processing>), and we discard reads with greater than 20 mismatches between forward and reverse reads or with a posterior Q score of less than 10. We use the wtNDM-1 samples as an estimate for error rates arising from the deep sequencing process, and we filter the non-selected libraries to remove variants with frequencies less than 2× of the expected frequency from sequencing noise alone, or variants with less than 5 reads. We then calculate the fitness scores for each variant in all conditions according to Eq. (1), where the frequency (freq) for a given variant (var) is used to calculate an enrichment ratio between the selected (sel) or non-selected (nosel) conditions, which is then normalized to the

enrichment ratio for the WT.

$$f_{mut} = \log_2 \left(\frac{\text{freq var sel}}{\text{freq var nosel}} \right) - \log_2 \left(\frac{\text{freq WT sel}}{\text{freq WT nosel}} \right) \quad (1)$$

All variants that pass filtering in the non-selected condition are considered to truly exist in the library, and if the same variants are not observed in conditions undergoing selection they are assumed to have been depleted by selection and are given a dummy count of 1 to simulate the lowest possible frequency. As for VIM-2, fitness scores for NDM-1 less than -4 are no fitter than non-sense variants, and we set fitness scores to a minimum of -4 for analyses.

Calculating mutational tolerance

To calculate DCA mutational tolerance, we compute the probability of each mutant at a given position, $P(\text{mut}, \text{pos})$, relative to the probability of the WT; their ratio is the logarithm of the model's "energy" between the mutant at a position and the wild-type, as $\Delta E = -\log[P(\text{mut}, \text{pos})/P(\text{WT})]$, such that we can define a normalized probability

$$P_{rel-mut,pos} = P(\text{mut}, \text{pos}) / \sum_{mut} P(\text{mut}, \text{pos}) = e^{-\Delta E} / \sum_{mut} e^{-\Delta E} \quad (2)$$

Note that because the sum is over all possible amino acids, including the WT, this quantity is independent of the specific WT residue. We then calculate the mutational tolerance using the relative probabilities of all mutants at that position through the Shannon Entropy.

$$\text{Mutational tolerance}_{DCA,pos} = - \sum_{mut=1}^{n=20} P_{rel-mut,pos} \log_2(P_{rel-mut,pos}) \quad (3)$$

For DMS mutational tolerance, we take an analogous approach to Firnberg et al.³ First, we normalize our fitness scores from a range of 0–1 relative to WT, where beneficial mutations are >1, and deleterious mutations range from 0 to 1. Since fitness scores of -4 or lower correspond to nonsense variants, we use -4 as the minimum range of fitness scores. We add a small padding factor of 10^{-5} to avoid complete zeroes.

$$f_{mut,norm} = \frac{(f_{mut} - (-4))}{0 - (-4)} + 10^{-5} \quad (4)$$

We use the proportion of normalized fitness relative to the sum of all normalized fitness scores as a measure of probability. If the position does not have synonymous mutants, WT is inserted with a normalized fitness of 1.

$$\text{Prop}_{mut,pos} = \frac{f_{mut,norm}}{\sum_{mut} f_{mut,norm}} \quad (5)$$

DMS mutational tolerance is then first calculated as the Shannon Entropy.

$$\text{Base mutational tolerance}_{DMS,pos} = - \sum_{mut=1}^n \text{Prop}_{mut,pos} \log_2(\text{Prop}_{mut,pos}) \quad (6)$$

We also scale the base mutational tolerance so that the effective number of amino acids is a maximum of 20, to account for positions which do not observe all 20 amino acids. The result is then the final

DMS mutational tolerance.

$$\text{Effective amino acids} = 2^{\text{Base mutational tolerance}_{\text{DMS,pos}}} \quad (7)$$

$$\text{Effective amino acids}_{\text{scaled}} = \text{Effective amino acids} \frac{20}{n} \quad (8)$$

$$\text{Mutational tolerance}_{\text{DMS,pos}} = \log_2(\text{Effective amino acids}_{\text{scaled}}) \quad (9)$$

Epistasis of specific mutations from DMS and DCA

In order to examine epistasis of individual mutations for the DMS data, we first isolate the reversion mutants from wtVIM-2 to NDM-1 (or vice versa) at positions with ‘differing’ WT residues (WT reversions, $N = 132$ after excluding 4 reversions not observed), and check their effects on fitness. As noted above, all fitness scores below -4 are set to a minimum of -4 . All reversions were tested for significant difference against a distribution of neutral effects, using a two-tailed z-test against the standard deviation of the synonymous distribution of each respective background, with false discovery rate (FDR) correction using the Benjamini–Hochberg procedure ($\alpha = 0.05$). Positions were considered incompatible if the effect was significantly negative in either or both backgrounds, and the remaining positions were labeled ‘compatible’, and treated the same as positions with ‘shared’ WT. Then, for shared and compatible positions, we compare the effect of all missense mutations ($N = 2392$) directly between the two backgrounds. A mutant is epistatic if the difference is significantly different by a two-tailed z-test against a null distribution with the same standard deviation as the NDM-1 distribution of synonymous mutants (greater standard deviation of the two backgrounds) with FDR controlled at $\alpha = 0.05$.

For the rest of the incompatible positions, positions were classified as ‘1-WT entrenched’ if the reversion is negative in only one background, and ‘2-WT entrenched’ if the reversion is negative in both. The reversion mutants were also analyzed by DCA using the calculated ΔE . Of note, DCA predicts many WT-swapping mutations to be beneficial compared to the DMS data (Fig. 4f, h). This may be a limit of the antibiotic selection assay, where a threshold like relationship between resistance and fitness for a given antibiotic concentration means variants with greater than needed resistance do not necessarily have higher fitness⁵, i.e., we have limited detection for gain of function. As a consequence, epistatic and non-epistatic mutations could be confounded for each other, as illustrated in Supplementary Fig. 6c. The 2-WT entrenched mutations are guaranteed by the experiment to be epistatic. However, for 1-WT entrenched mutations, if a reversion negative in one background is neutral in the other direction, it could be truly neutral (negative-neutral reversion, which is epistatic), or a beneficial effect that could not be detected (negative-positive reversion, which conforms to the non-epistatic anticorrelation in Fig. 4h). Still, there is reason to support the interpretation that those mutations are truly neutral in one background and deleterious in the other. First, in the distribution of effects observed in our previous analysis of VIM-2 variant EC_{50} (which unlike fitness, is not limited by the threshold effect) the rarity of gain-of-function mutations (1–2%) makes it improbable that there would be sufficient data points to create the anti-correlation expected in the absence of epistasis⁵. Moreover, as we have shown in Fig. 4i many 1-WT entrenched positions are statistically different from compatible ones in DCA, to the extent that some reach far away from the diagonal, just like 2-WT entrenched mutations.

Generation and dose-response assay of NDM single and combined mutants

To test if entrenched positions can compensate for mutations at nearby entrenched positions, the selected positions with entrenched WT behaviors were mutated in the NDM-1 background from the NDM-1

WT residue to the VIM-2 WT residue using Golden Gate cloning as single mutants (NDM-1 positions M67F, P68D, K125R, G197Y, A204L, K211Y, L218A, Y229W, I246V, I259L, A266V). Then, combinations of positions were generated by a second (or third for the triplet) round of mutations. Dose response curves were carried out to obtain the half-maximal effective concentration (EC_{50}) under AMP selection. For testing, mutants and wtNDM-1 were transformed into *E. coli* and grown overnight for 16 h at 30 °C, then diluted to a target inoculum of 1.2Mil cells/mL (OD_{600} of 0.0015) the next day. The diluted culture is grown for 1.5 h at 37 °C, then 180 μ L of culture is mixed with 20 μ L of 10 $\times$ AMP stock, with final ampicillin concentrations from 1 to 1024 μ g/mL. Growth under AMP selection is done for 6 h at 37 °C, and the OD_{600} of each culture is measured after selection. For each mutant or wtNDM-1, the OD_{600} across all selected AMP concentrations is plotted as a dose-response curve and fitted using a sigmoidal equation to obtain the EC_{50} .

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided as a Source Data file. Other data supporting the findings in this study are available in the paper or supplementary files. Raw deep sequencing data is available from the NCBI Sequence Read Archive (SRA) under Bioproject [PRJNA974578](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA974578). Previously published structures are available from the Protein Data Bank under accessions [5YD7](#), [3SPU](#), [4WD6](#), [6T5K](#), [6V3Q](#), [2YNT](#), [4UAM](#), [3L6N](#), [6T5L](#), [5MMD](#), [6JV4](#), [1BC2](#), [1M2X](#) and [1ZNB](#). Source data are provided with this paper.

Code availability

Specific software required for the analyses in this study are available publicly. Sequence space exploration for collecting B1 MBL sequences was conducted with MetaSSN⁶⁴ (<https://github.com/johnchen93/MetaSSN>). Deep sequencing analysis is performed with DMS-FastQ-processing⁶⁷ (<https://github.com/johnchen93/DMS-FastQ-processing>). DCA analysis is performed with adabmDCA²⁰ (<https://github.com/anna-pa-m/adabmDCA>).

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Acknowledgements

We thank members of the Tokuriki laboratory for insightful discussions and comments on the manuscript. We thank Dr. Eyal Akiva for their help in accessing the JGI IMG/M database. We thank the Canadian Institute of Health Research, Foundation Grant (FDN-148437, N.T.) for the financial

support. N.T. is a Michael Smith Foundation of Health Research career investigator.

Author contributions

J.Z.C. - conceptualization, software, experiments, data collection, analysis, methodology, writing and editing. M.B. - software, modeling, analysis, methodology, writing and editing. D.L. - experiments, analysis and editing. S.C. - software, modeling, analysis, methodology, writing and editing. F.Z. - supervision, methodology and editing. M.W. - supervision, methodology and editing. N.T. - conceptualization, supervision, funding acquisition, project administration and editing.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-024-52614-w>.

Correspondence and requests for materials should be addressed to N. Tokuriki.

Peer review information *Nature Communications* thanks Yeonwoo Park and the other, anonymous, reviewers for their contribution to the peer review of this work. A peer review file is available.

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